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Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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Oil compressibility

1. Introduction

Fluid compressibility is a physical property that describes the extent to which a fluid can be compressed or expanded under changes in pressure. Furthermore, knowledge of this property can provide insight into the composition of the components that make up a mixture, which in our particular case is a hydrocarbon sample. The composition of hydrocarbons has a direct influence on their compressibility. In general, the greater the proportion of light hydrocarbon components, the more compressible the fluid becomes. Conversely, mixtures containing a higher fraction of heavy components are less compressible due to the larger size and lower mobility of their constituent molecules.

In liquid hydrocarbon systems, the liquid compressibility, c_o , varies with pressure because crude oil contains dissolved gas that remains in solution while the pressure is above the bubble-point pressure, p_b . When the pressure falls below p_b , the dissolved gas begins to evolve from the oil, resulting in gas liberation and a reduction in the liquid phase mass. This phase behavior causes the pressure-dependent variation observed in the oil compressibility.

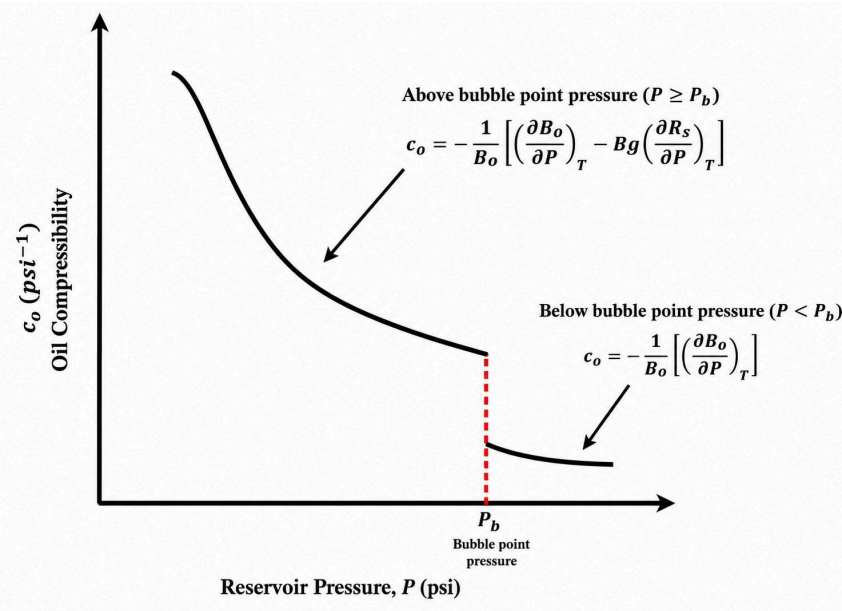
$p > p_b$

$$c_o = -\frac{1}{B_o} \left(\frac{\partial B_o}{\partial p} \right)_T$$

$p < p_b$

$$c_o = -\frac{1}{B_o} \left[\left(\frac{\partial B_o}{\partial p} \right)_T + B_g \left(\frac{\partial R_s}{\partial p} \right)_T \right]$$

The following figure illustrates the overall behavior of oil compressibility, c_o , as a function of pressure. A discontinuity occurs at the bubble-point pressure due to the liberation of dissolved gas from the oil, which causes a sharp increase in the value of c_o . In addition, below the bubble-point pressure (P_b), the oil undergoes a loss of mass as gas evolves from the liquid phase. Consequently, an analysis of the oil formation volume factor (B_o) reveals that the same value of B_o may correspond to two different pressure conditions. This behavior is explained by the combined effects of oil compression above the bubble point and oil expansion associated with gas liberation below the bubble-point pressure.



There are several numerical correlations available to compute the oil compressibility, c_o , some of the most widely used include:

- Vazques and Beggs
- Petrosky and Farshad
- Kartoadmodjo and Schmidt

2. Numerical solution

```
$ p>p_b $  
$$ c_o=\frac{-1433+5R_s+17.2T-1180\gamma_{gc}+12.61\gamma_{API}}{p\times 10^5} $$  
$ p<p_b $  
$$ \ln(c_o) = -7.114-1.394\ln p-0.981\ln T+0.770\ln(\gamma_{API}) $$
```

2.1 Import libraries

```
In [1]: import numpy as np  
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [22]: gamma_g=0.9  
gamma_o=31  
T=np.array([100,110,120])
```

```
pb=2200
Rsb=300

p=np.linspace(500,3000,100)
```

2.3 Calculation and plotting

Note: To calculate the oil factor volume factor B_o is required to know the solution gas-oil ratio R_s . Please review the corresponding notebook

```
In [36]: plt.figure(dpi=300, figsize=(8,4))

for i in range(len(T)):

    co_1 = []
    co_2 = []

    p_1 = []
    p_2 = []

    for j in range(len(p)):

        # co at pressure lower than pb
        if p[j] <= pb:
            co1=np.exp(-7.573-1.450*np.log(p[j])-0.383*np.log(pb) +1.402*np.log(T[i]+460)+0.256*np.log(gamma_o)+0.449*np.log(Rsb))

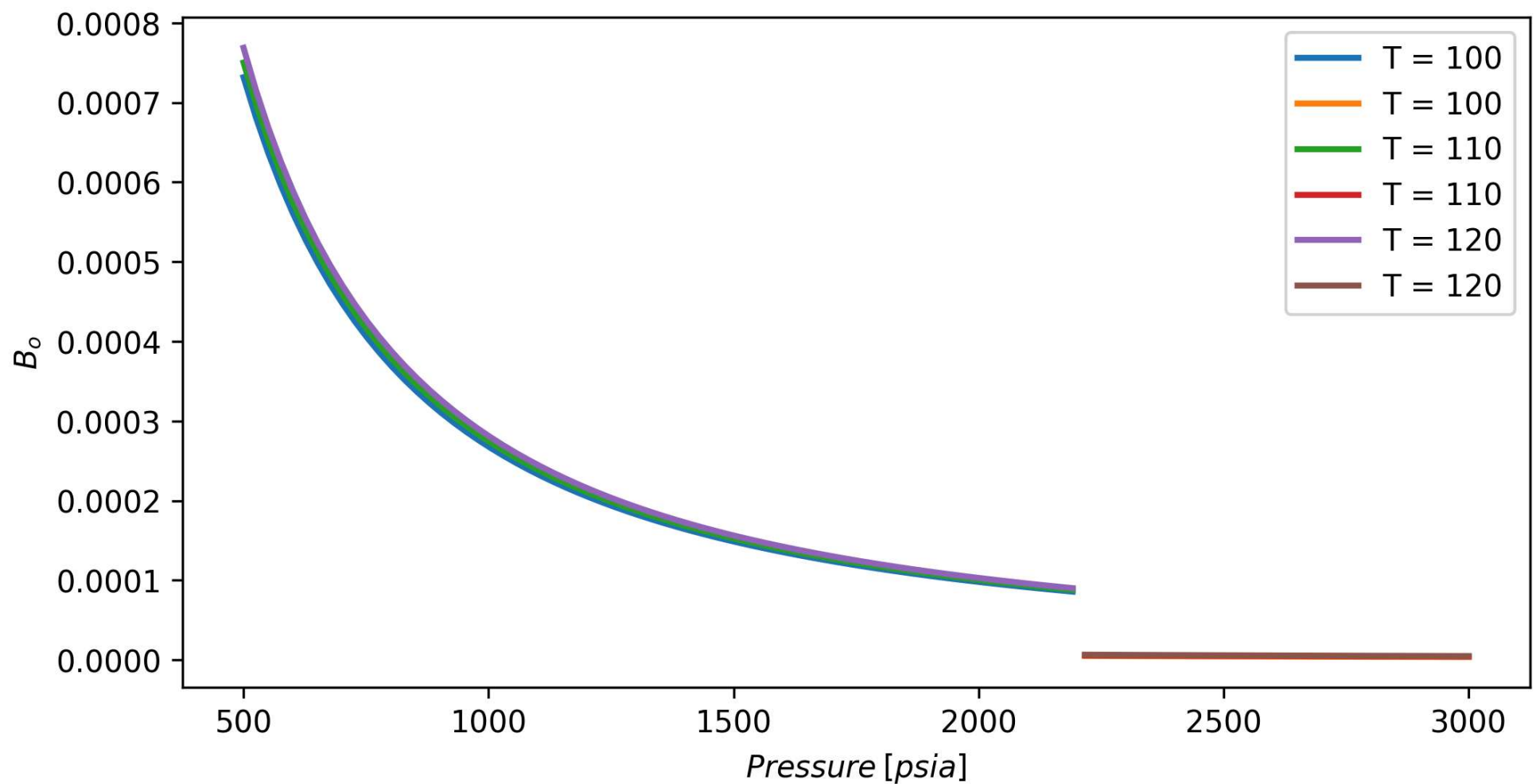
            co_1.append(co1)
            p_1.append(p[j])

        # co at pressure greater than pb
        if p[j] >= pb:
            co2 = (-1433+5*Rsb+17.2*T[i]-1180*gamma_g+12.61*gamma_o)/(p[j]*10**5)

            co_2.append(co2)
            p_2.append(p[j])

    plt.plot(p_1, co_1,p_2, co_2, linestyle='-', linewidth=2,
             label=f'T = {T[i]}')

plt.xlabel('$Pressure\:[psia]$')
plt.ylabel('$B_o$')
# plt.grid(True)
plt.legend()
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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In []: